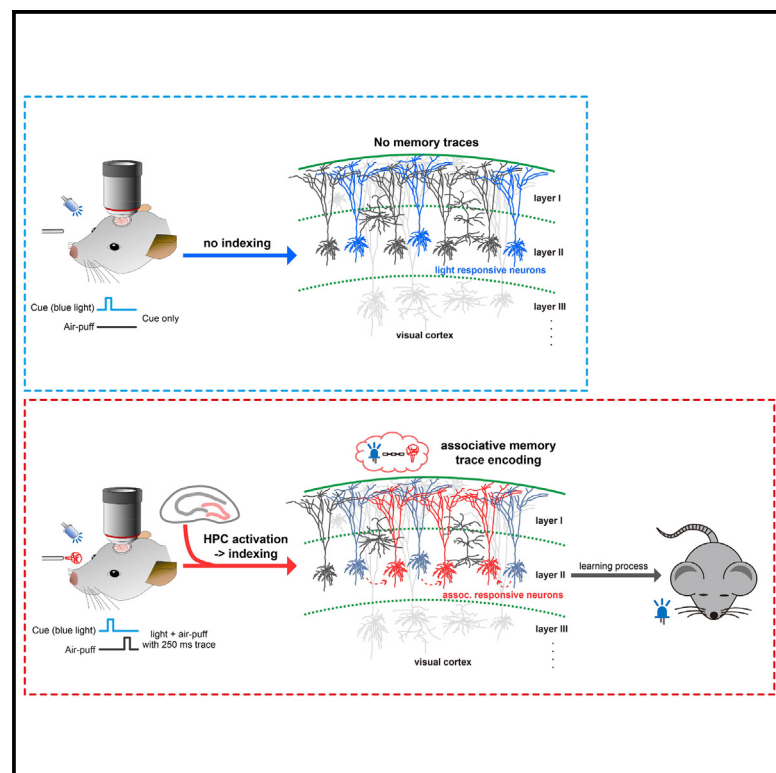


Hippocampus alters visual representation to encode new memory

Graphical abstract



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In brief

Liu et al. demonstrate that associative learning induces changes of the active neuronal ensemble in mouse visual cortex. The study reveals that hippocampal activity plays an indexing role in forming these altered representations, which are crucial for associative memory processes.

Highlights

- A neuron ensemble in layer II of the mouse visual cortex (VIS) in trace eyeblink conditioning
- VIS responds to paired stimulus but not discrete stimuli, resembling associative learning
- Hippocampal activity is crucial for altering representations in the VIS as an index
- Fos⁺ neurons in VIS are hubs for maintaining associative memory



Article

Hippocampus alters visual representation to encode new memory

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SUMMARY

Hippocampal-cortical interaction is crucial for episodic memory encoding and retaining, but when and how this interaction occurs remains elusive. Here, within the trace eyeblink conditioning paradigm, we found a neuronal ensemble in layer II of mouse visual cortex (VIS) that responded to the paired stimulus of light flash (cue, or conditioned stimulus [CS]) and air puff (unconditioned stimulus [US]), but not discrete stimuli, resembling an associative event during learning. This neuronal representation in VIS is dependent on the hippocampus and contributes to encoding the association. Optogenetic activation of hippocampus can promote the emerging representation to allow the association of separated cues. Mechanistically, *fos*⁺ engram cells, modulated by *VIP*⁺ neurons, are hubs of the association-activated ensemble in VIS. The hippocampus-modulated memory ensemble emerging in VIS at an early stage implicates the structural organization of the memory network to encode new memory.

INTRODUCTION

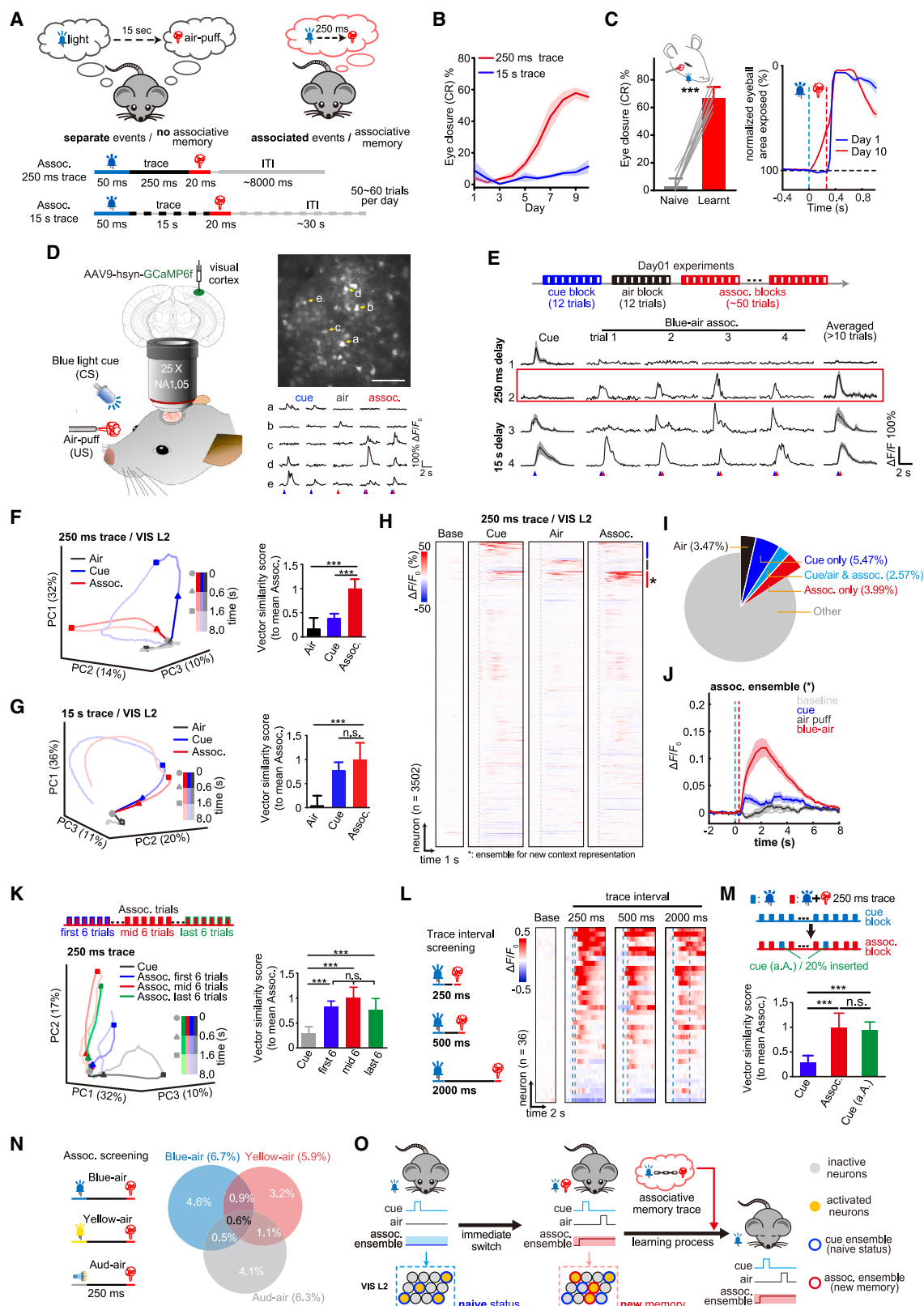
Although synaptic plasticity underpins memory storage,^{1,2} how memories are encoded and retained in the neuronal network remains elusive. In the mammalian brain, hippocampus (HPC) critically mediates contextual/declarative memory encoding.^{3–5} Meanwhile, the memory traces or engram cells have also been found in the cortical regions, such as retrosplenial cortex (RSP), cingulate cortex (ACC), medial prefrontal cortex (mPFC), and sensory neocortices in various stages of memory processes.^{6–11} The hippocampal memory indexing theory posits that HPC creates the indices of distributed neocortical activation patterns through synaptic plasticity mechanisms during memory encoding.^{12,13} In the process of memory retrieval, the hippocampal index is represented by the subset of the initial cortical input to convey the activation of the entire memory recollection. This theoretical framework has been supported by clinical and experimental observations,^{13,14} but the regulatory details of the hippocampal-neocortical pathway during memory formation lack empirical support.

Eyeblink conditioning is a widely utilized associative learning paradigm for studying declarative memory processes.^{15–17} This approach pairs a neutral conditioned stimulus (CS), like auditory, visual, or whisker stimulation, with an unconditioned stimulus (US), usually an air puff to the eye, which elicits a reflexive eyelid closure (unconditioned response [UR]).^{15,18,19} After training, animals close their eyes in response to the CS, known

as the conditioned response (CR). Two main paradigms are used: delay and trace. In delay conditioning, the CS is presented in advance and lasts until the spontaneous termination with the US. Cerebellum is key for the CR in delay conditioning.^{15,16} Conversely, trace conditioning has a blank interval between the CS and US. HPC is crucial for acquiring the CR in trace conditioning, and neocortices are also involved, sharing plenty of features and mechanisms with the episodic memory process.^{15,18–22} Thus, trace eyeblink conditioning serves as a model to study memory encoding and associative learning involving hippocampal-neocortical interactions.

To reveal how the memory traces are encoded in neocortex and how the hippocampal-neocortical circuit functions in memory formation at the neuronal level, we recorded the transient calcium (Ca^{2+}) events with the genetically encoded Ca^{2+} indicator GCaMP6f of mouse visual cortex (VIS) during the visual-cued trace eyeblink conditioning paradigm. In the 250 ms trace interval training, we found the emergence of a neuronal ensemble in layer II of VIS that specifically responded to the CS-US paired stimuli but not to the individual CS or US stimulus, representing the association memory traces. The emerging ensemble was established within several trials of efficient pairing of the CS and US, much faster than the rising of the CR level. These neocortical memory traces are regulated by HPC, as the chemogenetic inhibition of dentate gyrus (DG) in HPC diminished the generation of the associative ensembles in VIS at the early stage of trace conditioning. Consistently, the activation of DG led to the shift of





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neuronal responding patterns in VIS. These observations reinforced the HPC indexing theory with empirical support. To investigate how the associative memory traces are encoded at the beginning of the trace conditioning, we labeled the *fos*⁺ cells (engrams) in VIS through targeted recombination in active populations (TRAP) and found an increased percentage of labeled engrams for specific associative events with long-lasting activities. Optogenetic activation of *fos*⁺ cells in VIS increased the CR level, and chemogenetic inactivation of the engrams altered the entire activation patterns of VIS in the trace conditioning. At the local circuit level, we found that vasoactive intestinal peptide-positive (VIP⁺) inhibitory neurons, but not parvalbumin (PV⁺) or somatostatin (SOM⁺) neurons, selected the neuronal ensemble for the association in layer II of VIS. In summary, our research on the trace eyeblink conditioning illustrates that the memory traces in sensory neocortex engage in the associative learning process with the hub of engram cells and that these traces are HPC-activation dependent, reinforcing the indexing role of HPC during memory formation.

RESULTS

Associative learning alters cortical representation to the visual cue in layer II of mouse VIS

To investigate neural responses in VIS during associative learning, we employed a trace conditioning paradigm in head-fixed mice, pairing a 50 ms blue light (as the cue or CS) with a 20 ms air puff (US) at intervals of 250 ms (short trace) or 15 s (long trace) (Figure 1A). After 10 days of training (60 trials/day), short-trace mice developed conditioned eyeblink responses (CRs) in over 60% of trials within 300 ms post-CS (Figures 1B and 1C), whereas long-trace mice and naive controls (day 1) showed no increased CRs (Figures 1B and 1C), indicating the associative memory formation exclusively in the short-trace group.

During the behavioral training phase, we used two-photon calcium imaging to record neural activity in mice expressing GCaMP6f in layer II of VIS, both in response to isolated stimuli and during associative training sessions (Figures 1D and S1A–S1E). On day 1, the first day of training, the mice were subjected to 12 trials of flash stimulation (CS, 50 ms, referred to as the cue block) and 12 trials of air puff (US, 20 ms, referred to as the air block) prior to the commencement of paired stimulus training (CS-US, referred to as the association block) (Figure 1E). It is notable that in the association trials, although the CS was presented and mice did not close their eyes to light stimuli on day 1, some responding neurons in the cue block were not activated. Instead, an emerging group of neurons, inactive in cue blocks, were activated by light in these trials (Figure 1E). This was specific to the 250 ms trace group, not the 15 s trace group, suggesting that associative training may recruit a distinct set of neurons in VIS for the association scenario, as opposed to the individual stimuli presented in isolation.

The distinct neural responses were observed not only in individual neurons but also at the population level. Principal-component analysis (PCA) and the vector similarity scores (see STAR Methods and Figures S1F–S1H and S2 for details) of the neural activity in VIS revealed distinct population trajectories between cue-only and paired trials in the 250 ms trace group during initial training (Figure 1F), with separation beginning at light stimulus onset. This divergence contrasted with the 15 s trace group's overlapping trajectories (Figure 1G) and differed from the linear summation of isolated CS/US responses (Figures S2E and S2F), indicating that the neural representation in VIS for the light was altered to reflect the paired-trials scenario rather than the individual stimuli. The activity heatmaps confirmed three segregated neuronal subgroups activated during CS-only, US-only, and paired trials (Figure 1H). The paired stimuli activated a small subset of neurons (3.99%; Figure 1I), which showed minimal activities in isolated trials (Figure 1J). This kind of neuron subset

Figure 1. Learning immediately alters neuronal representation to visual cues in layer II of the mouse visual cortex

- (A) Trace eyeblink conditioning paradigms (250 ms/15 s trace intervals).
 (B) Learning curves for CR during the associative learning ($n = 11$ mice for 250 ms trace, $n = 7$ mice for 15 s trace). Mean $\pm$ SEM.
 (C) Pre-/post-learning CR (%) and normalized eyeblink exposure ($n = 10$ mice, paired t test, $p = 2.59 \times 10^{-5}$). Left: mean $\pm$ SD. Right: normalized eyeblink exposed (%), used to depict the extent of the mouse's eyelid closure in 300 ms response window ($n = 48$ trials). Mean $\pm$ SEM.
 (D) Calcium imaging in layer II of VIS. Example neuron activities during 250 ms trace conditioning (day 1). Stimulus onset markers: blue (CS of light flash), red (US of air puff), and the same hereafter. Mean $\pm$ SEM.
 (E) Top: the mice were stimulated by blue-light cue, air puff, and association trials sequentially. Bottom: calcium traces of four neurons during the paired trials. Mean $\pm$ SEM.
 (F and G) The PCA trajectories and vector similarity scores for layer II neurons of VIS during 250 ms (F, $n = 2,428$ neurons from 12 mice) and 15 s (G, $n = 2,614$ neurons from 7 mice) trace paradigms. Welch's t tests for the bar plot. Cue versus association: $p = 4.67 \times 10^{-13}$ in (F) and $p = 0.079$ in (G); air versus association: $p = 1.31 \times 10^{-9}$ in (F) and $p = 5.50 \times 10^{-20}$ in (G). Mean $\pm$ SD.
 (H) Heatmaps categorizing VIS layer II neurons into five response clusters (250 ms trace; $n = 3,502$ neurons from 11 mice). Dashed lines: stimulus onset.
 (I) Pie chart of stimulus-responsive neuron ratios (250 ms trace, day 1). In this pie chart, the neurons that were activated by both the CS or US and the paired trials are marked in light blue to show the divergence between the discrete trials and the paired trials.
 (J) The averaged activity of association-activated neuronal ensembles (250 ms trace, day 1). Vertical lines denote stimulus onset (blue: CS of light flash; red: US of air puff, and the same hereafter). Mean $\pm$ SEM.
 (K) The PCA trajectories and vector similarity scores across early/middle/late paired trials in the 250 ms trace eyeblink conditioning on day 1 ($n = 2,428$ neurons from 12 mice). Welch's t tests for the bar plot: cue versus first 6 trials, $p = 0.0150$; cue versus middle 6 trials, $p = 9.55 \times 10^{-5}$; cue versus last 6 trials, $p = 2.78 \times 10^{-3}$; first 6 trials versus middle 6 trials, $p = 0.388$; first 6 trials versus last 6 trials, $p = 0.659$; and middle 6 trials versus last 6 trials, $p = 0.736$. Mean $\pm$ SD.
 (L) Heatmaps of neurons activated by 250/500/2,000 ms trace intervals ($n = 36$ cells from 3 mice).
 (M) Vector similarity scores for cue, association, and cue (a.a.) in 250 ms trace trials on day 1. Welch's t tests for the bar plot: cue versus association, $p = 1.41 \times 10^{-13}$; cue versus recall, $p = 2.89 \times 10^{-7}$; and association versus recall, $p = 0.459$. Mean $\pm$ SD.
 (N) Venn diagram of neuron response overlaps across association stimuli (250 ms trace, day 1).
 (O) Schematic of neuronal responses during associative conditioning in layer II of VIS.

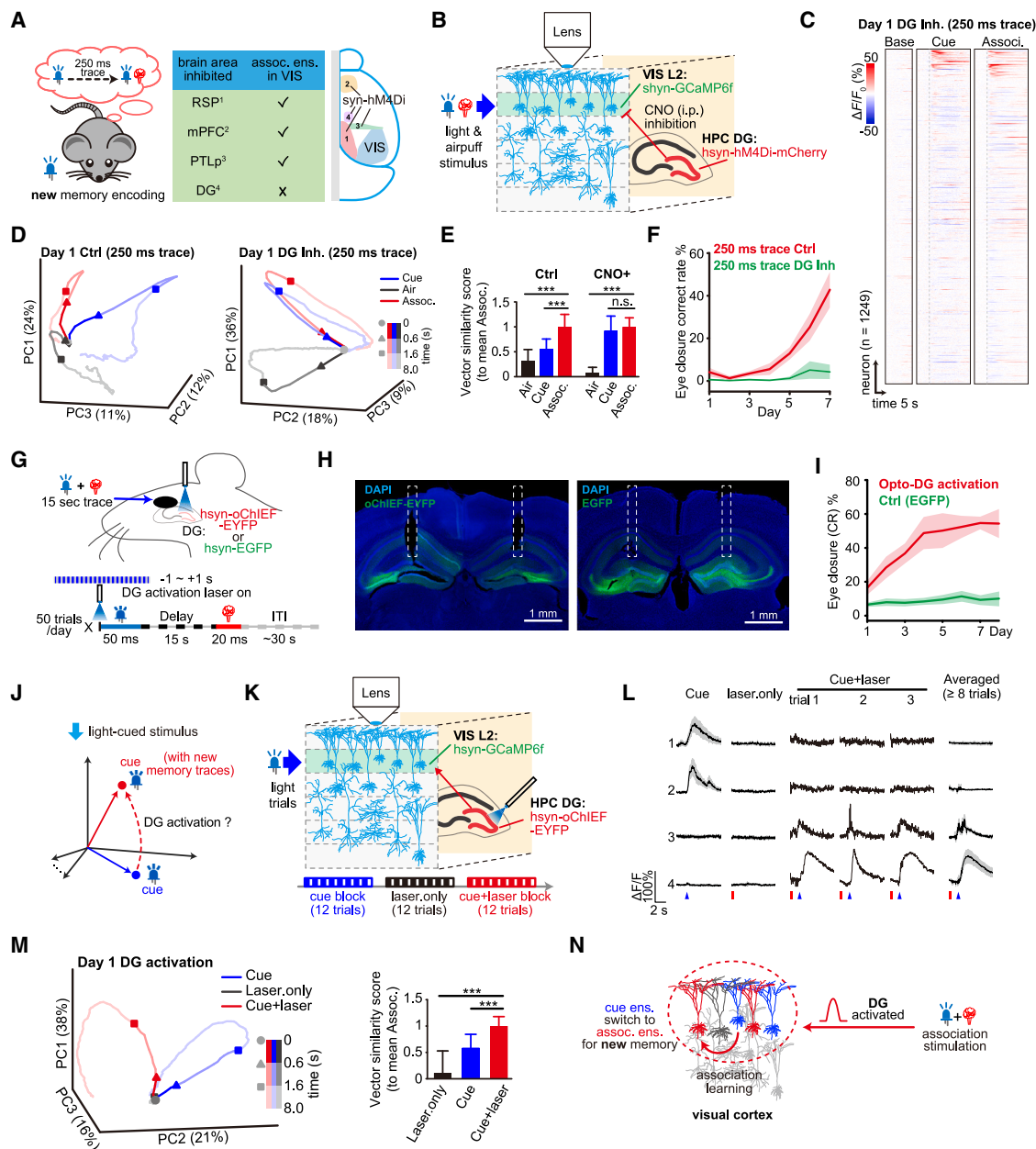


Figure 2. Hippocampal activity alters visual representation during associative learning

(A) The brain region inhibition screening for the emerging association-activated ensemble in layer II of VIS.
(B) The diagram for the DG inhibition experiment.
(C) The heatmaps of the neuronal response heatmaps to cue/association stimuli during DG inhibition in layer II of VIS (250 ms trace; $n = 1,249$ cells from 6 mice).
(D) The PCA trajectories of layer II neurons of VIS with/without DG inhibition during 250 ms trace conditioning.
(E) The vector similarity score of layer II neurons of VIS in response to cue, air puff, and blue-air association stimuli in the 250 ms trace eyeblink conditioning on day 1 without and with DG inhibition. Welch's t tests for the bar plot: for the control (Ctrl) group, cue versus association, $p = 9.72 \times 10^{-5}$ and air versus association, $p = 4.27 \times 10^{-7}$; for the CNO⁺ group, cue versus association, $p = 0.92$ and air versus association, $p = 9.51 \times 10^{-4}$. Mean $\pm$ SD.
(F) The behavior comparison between the DG inhibition group and the control group. ANOVA test: $p = 6.67 \times 10^{-4}$ for days 5–7, $n = 4$ for the DG inhibition group, and $n = 5$ for the control group. Mean $\pm$ SEM.
(G) Bilateral DG optogenetic activation schematic.
(H) DG viral expression verification (oChIEF versus EGFP). Scale bar: 1 mm.
(I) The correct rate of eyelid closure with/without DG activation ($n = 3$ animals for the oChIEF group and $n = 5$ animals for the EGFP, 80 trials per day). Mean $\pm$ SEM.
(J) The proposed DG-mediated modulation of VIS ensemble dynamics in PCA space.
(K) Unilateral DG optogenetic activation design enabling simultaneous imaging.

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was not present in the 15 s trace trials (Figure S3), suggesting that these "associative memory ensemble" neurons may specifically represent light-air puff associations. We then further explored their properties in the visual representation.

Firstly, this associative ensemble formed rapidly. We found that PCA trajectories of the association block diverged from the cue block within the first six paired trials (Figures 1K and S2G), while repeated cue trials maintained stable patterns (Figures S2H and S2I), indicating that adaptation barely affected this ensemble activation changing. Secondly, this representation showed temporal flexibility, that is, varying inter-stimulus interval traces (250–2,000 ms) consistently activated the same neuronal subset (Figure 1L) with stable averaged activity profiles (Figures S4F–S4H). Thirdly, the altered visual representation remained intact even when the US was omitted. In mice trained with the 250 ms trace, we introduced 20% cue-only trials (indicated as cue (a.A.) [cue trial after association]) and found that the neural responses to the cue (a.A.) trials were similar to the association block and distinct from the cue block (Figures 1M, S2A, and S2D). This reinforces that the neural activation pattern is tied contextually to the associative scenario. Lastly, the emerging associative ensembles showed stimulus specificity, as different CS-US pairs (blue/yellow light-air puff or tone-air puff) activated neuronal populations with small overlap (Figure 1N) with distinct PCA trajectories (Figures S4A–S4E), demonstrating the precise coding of specific associations in layer II of VIS.

Thus, associative training reorganizes VIS layer II activity through a specialized ensemble that emerges rapidly, maintains temporal and contextual fidelity, and specifically encodes paired stimuli. These enduring activity patterns may create perceived simultaneity between discrete events, facilitating synaptic integration for memory formation (Figure 1O).

Hippocampal activity during learning alters the visual representation in VIS

To investigate how associative stimulation alters visual representations, we examined hippocampal-cortical interactions given prior evidence linking cortical memory formation to hippocampal inputs.²³ Using designer receptors exclusively activated by designer drugs (DREADD)-based inactivation across multiple brain regions, including mPFC, RSP, posterior parietal cortex (PTLP), and HPC,²⁴ we found that only hippocampal suppression prevented the alternation of visual representations in VIS to a large extent during association training (Figures 2A and S4I–S4K). In the experiment, AAV9-hsyn-hM4Di-mCherry was injected in DG and clozapine-N-oxide (CNO; 2 mg/kg, intraperitoneal [i.p.] 60 min before training) was injected before the association training (Figures 2B and S5A). We found that the neural responses to the association block overlapped substantially with the cue trials with HPC inactivation (Figure 2C), and the PCA trajectories failed to differentiate between the cue trial and the association trial (Figures 2D and 2E) in the 250 ms trace group. Critically, preserved air-puff-evoked blinks confirmed

that this effect was not caused by motor or sensory changes. These results establish that hippocampal activity, particularly in DG, is required to alter visual representation in the early phase of trace conditioning.

Consistent with the notion that the altered visual representation is required for associating temporally separated stimuli during learning (Figure 1O), we found that inhibiting DG blocks associative learning. HPC-inhibited mice showed no altered visual representation in VIS and no increased eyeblink behavior after 7 days of conditioning (Figure 2F). Thus, the altered visual representation was closely linked to the association of temporally separated stimuli.

To determine whether introducing the alternation of visual representation facilitates associative learning of the stimuli that are temporally distant (i.e., 15 s trace interval), we injected AAV9-hsyn-oChIEF-EYFP into DG bilaterally (Figures 2G and 2H). The mice underwent 15 s trace conditioning with optogenetic DG activation (20 Hz, 5 ms on and 45 ms off in cycle) applied for 2 s at light cue onset (–1 to +1 s). Despite lacking explicit information about the association, optogenetic DG stimulation enabled mice to associate the light with the air-puff stimuli. The mice exhibited progressive increases in light-triggered eyeblink behavior over 8 training days (Figure 2I). Temporal analysis revealed sustained eye closure throughout the 15 s interval in the mice, indicating that the animals learned the association between the CS of light and the US of air puff but not the exact air-puff timing (Figures S5B and S5C). After learning, mice showed increased CR events in response to the CS of light without laser stimuli in DG (Figures S5D and S5E). In contrast, CR behavior was not observed in the control group (EGFP instead of oChIEF), which also received laser stimuli in DG.

We next examined whether hippocampal (DG) activation alters visual representations in VIS (Figure 2J). The neural activities in layer II of VIS were recorded with or without optogenetic activation of DG (AAV9-hsyn-oChIEF-EYFP, –1 to +1 s at onset of light cue, 20 Hz). DG activation alone did not evoke VIS responses (Figure 2L). However, simultaneous DG activation and light exposure significantly altered the neural response: some original light-responsive neurons became silent, while a distinct population was activated (Figure 2L). PCA revealed nearly orthogonal population activity trajectories between the conditions with versus without DG activation (Figure 2M), indicating the shift of visual representation with DG activation. These observations suggest that the altered visual representation, closely tied to the association learning of two temporally separated cues, is generated in VIS by both sensory input and the long-range effect of HPC activation (Figure 2N).

Activity level of association-activated ensembles modulates learning speed

To determine whether altered VIS representations correlate with learning rates, mice were categorized into fast (>50% eyeblink) and slow (<50%) learners based on their performance on day

(L) The calcium fluorescence signal recording of four neurons in the cue, laser-on only, and cue+laser-on trials. Mean $\pm$ SEM.

(M) The PCA trajectories and the bar plot of vector similarity score of layer II neurons of VIS in response to cue, laser-on only, and cue+laser-on trials on day 1 ($n = 469$ cells). Welch's t tests for the bar plot: cue versus cue+laser on, $p = 3.74 \times 10^{-6}$ and laser-on versus cue+laser on, $p = 1.25 \times 10^{-23}$. Mean $\pm$ SD.

(N) The working model of DG's long-range regulatory role in VIS association ensembles.

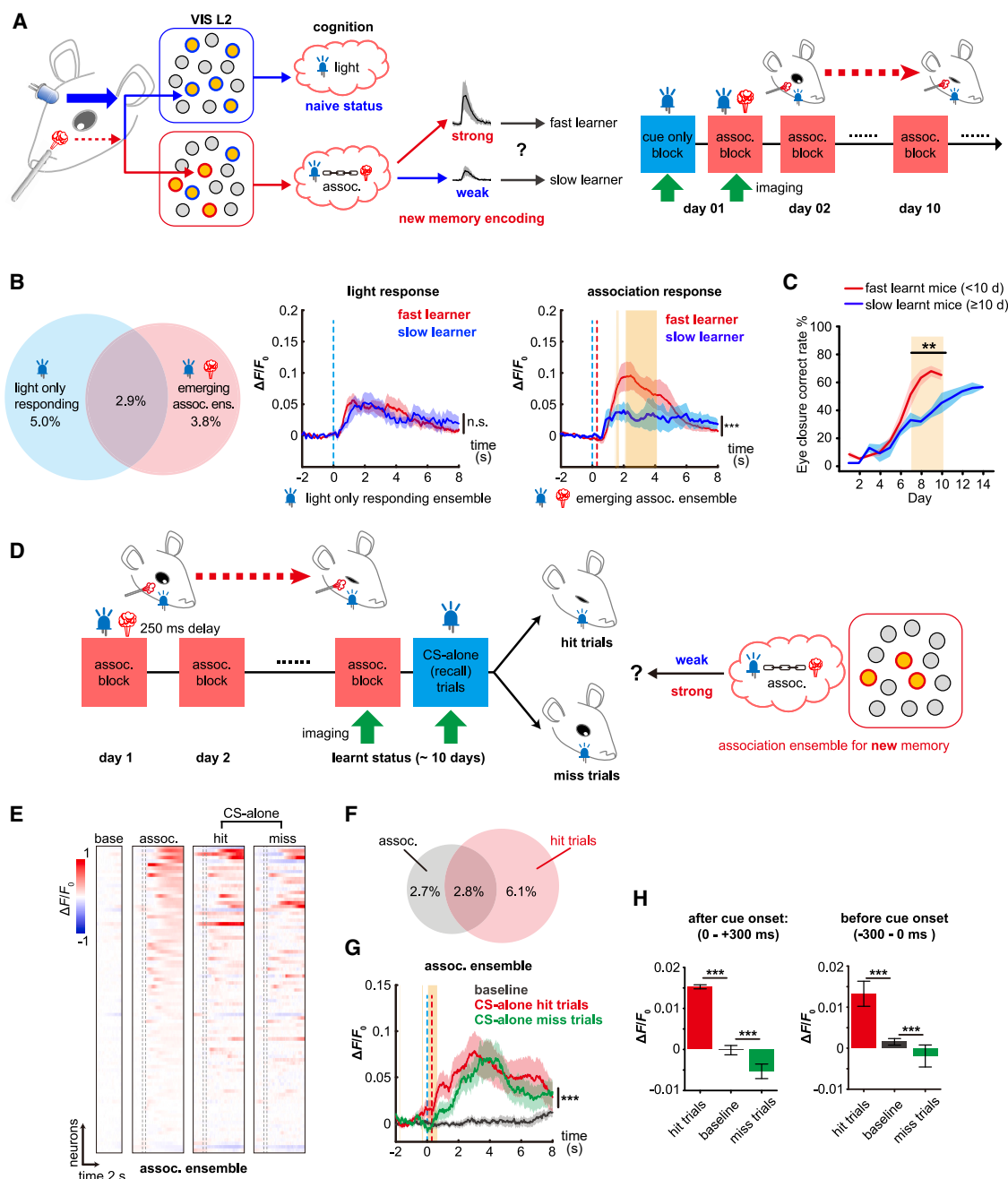


Figure 3. The associative memory representation in VIS affects learning and memory recall

(A) The diagram of the hypothesis on the altered neuronal representation in VIS on day 1 and the associative learning progress.

(B) The responding intensities of the cue responding and the emerging association-activated ensembles on day 1. Yellow shading: time windows with significant fast/slow learner differences (Mann-Whitney U test, $p < 0.05$). Two-way ANOVA test for the comparisons between the averaged curve: the averaged light responses of the light-only responding neurons, $p = 0.48$, $n = 44$ for the fast learner group, and $n = 38$ for the slow learner group; the averaged association responses of the emerging association-activated neurons, $p = 1.38 \times 10^{-61}$, $n = 28$ for the fast learner group, and $n = 19$ for the slow learner group. The p value is for the main effects and the same hereafter. Mean $\pm$ SEM.

(C) Behavioral divergence: fast versus slow learners. Yellow shading: days with significant eye closure correct rate differences (two-way ANOVA test, $p < 0.01$). Mean $\pm$ SEM.

(D) The diagram shows the hypothesis that the association-activated ensemble impacts the recall behavior of the mice.

(E) The heatmaps of the responses of the association-activated ensemble to the association and recall trials ($n = 117$ cells from 9 animals).

(F) The Venn chart shows the overlapping of the association responding and the CS-alone hit trial responding neurons.

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10 (Figure 3A). Neuronal ensembles selectively activated by light (cue blocks) or paired cues (association blocks) were identified (Figures 3B, S6A, and S6B). While light-activated ensembles showed comparable activity between the groups during early training (Figures 3B, S6C, and S6E), association-activated ensembles exhibited significantly higher activity in fast learners (Figures 3B, S6D, and S6F). Behaviorally, fast learners displayed steeper learning curves from mid-training stages compared to slow learners (Figure 3C).

Activity level of association-activated ensembles contributes to successful recall of memory

To investigate whether altered visual representations, particularly association-activated neuronal ensembles, correlate with memory recall, we analyzed the relationship between the activity of specific neuron ensembles and eyeblink behavior on day 10 after training. The mice underwent association blocks interspersed with 20% CS-alone recall trials (no US), categorized as "hit" (eyeblink ≤ 300 ms post-blue-light CS) or "miss" (no blink) (Figure 3D). We found that neurons active during CS-alone trials overlapped with those responsive in association blocks (Figures 3E and S7A). A significant proportion of association-activated neurons were recruited during hit trials (Figures 3F and S7B). Calcium signals in these neurons were markedly stronger in hits versus misses (Figures 3G and S7C), particularly during the early phase (0–300 ms) (Figure 3H). Notably, in miss trials, these ensembles showed reduced activity during the early cue onset stage compared to baseline (Figure 3H). Thereby, elevated activities in the association-activated ensemble in VIS, especially in the early stage of recall, were correlated with the successful retrieval of the memory.

We also observed dynamic changes in VIS visual representations across training days. Some neurons highly responsive during day 1 association blocks lost responsiveness by day 10 (Figures S7A and S7B). Among the neurons that were newly activated in association blocks on the first day (representing 3.6% of all recorded neurons), 21% of them were still activated in the hit trials on day 10 (Figure S7D). This persistent subset exhibited significantly stronger activity in hit versus miss trials, particularly during the early cue phase (Figures S7E–S7G). These neurons also showed higher activity in day 10 hit trials compared to day 1 (Figure S7G). While association-altered representations underwent continuous changes over time, a stable component persisted. These findings indicate that VIS association-activated ensembles serve as memory traces for the light-air puff association.

TRAP-labeled fos⁺ cells of association-activated ensemble in VIS-evoked memory recall

To establish causality between association-activated neurons and memory retrieval, we used the TRAP method to label and

manipulate neurons activated during learning.^{25,26} The Fos-creER^{T2} mice were injected with AAV9-DIO-EF1 α -oChIEF-mCherry in layer II of VIS (Figure 4A). After 10 days of blue light-air puff (blue-air) training, mice underwent 50 additional association trials 24 h after tamoxifen (TM) injection (i.p., 150 mg/kg) to label the responsive fos⁺ neurons during association stimuli in learned mice. On day 17, optogenetic reactivation (473 nm laser: 20 Hz, 5 ms/pulse, 10 pulses/trial, 10–13 s inter-trial interval in 3 min blocks) of labeled neurons without visual cues elicited significantly more eyeblink behaviors during laser-on versus laser-off trials (Figure 4B). The eye closure occurred approximately 1–4 s after the laser onset (Figure 4C), slower than blue-light-triggered responses (<300 ms), suggesting memory retrieval rather than direct cue-driven behavior. Notably, activating the fos⁺ neurons labeled on day 1 also induced eyeblink behavior (Figure S8A), confirming the association-activated fos⁺ cells as memory traces.

TRAP-labeled fos⁺ cells were the hub neurons of the altered visual representation during association training

To determine whether TRAP-labeled fos⁺ neurons (presumably memory engrams) are indeed the association-activated neuron ensembles identified via Ca²⁺ activities, we injected both AAV9-hsyn-GCaMP6f to monitor neuronal activity and AAV9-DIO-EF1 α -hM4Di-mCherry to label the neurons activated during blue-air association training in the Fos-creER^{T2} mice on day 1, with TM injection (Figure S8B). On day 8, imaging during 250 ms trace trials with three pairings—blue-air, blue light-yellow light, and blue light-audio (blue-audio)—revealed that 17.1% (176/1,029) of GCaMP6f-expressing neurons were TRAP labeled, while 8.3% (85/1,029) responded to the blue-air pairing. Among these blue-air-responsive neurons, 21% were TRAP labeled (Figures 4E and 4F). The TRAP-labeled neurons included the cells non-specific to blue-air training. Non-TRAP-labeled (fos[−]) neurons showed comparable responding proportion across all pairings (Figure 4G), whereas TRAP-labeled (fos⁺) neurons exhibited higher responsiveness to blue-air trials (5.7% versus 3.6% in fos[−] neurons; asterisks in Figure 4G). The fos⁺ ensemble also contained a greater proportion of cells selectively responsive to blue-air trials than other pairings (Figure S8D). Additionally, fos⁺ neurons showed stronger Ca²⁺ signals during blue-air trials than blue-audio trials, while fos[−] neurons displayed similar responses to both (Figure S8E). Although TRAP-labeled neurons were not exclusively restricted to the blue-air association-activated ensemble, they showed a specific proportional preference.

We compared Ca²⁺ activities between TRAP-labeled and non-labeled association-activated ensembles. TRAP-labeled neurons exhibited sustained Ca²⁺ activity, significantly exceeding fos[−] neurons during 3–6 s after the paired stimuli (Figure 4H). We also found that the fos⁺ neurons maintained prolonged activation compared to fos[−] cells during paired trials on the labeling

(G) Ensemble activation in CS-alone hit versus miss trials. Yellow shading: significant periods (Wilcoxon signed rank test, $p < 0.05$). Two-way ANOVA test for the comparisons between the averaged curves: $p = 5.15 \times 10^{-148}$ and $n = 88$ neurons. Mean $\pm$ SEM.

(H) The area under curve (AUC) of ensemble activity during baseline/hit/miss trials. One-way ANOVA test for the bar plot: after cue onset group, hit trials versus baseline, $p = 1.58 \times 10^{-33}$; miss trials versus baseline, $p = 8.73 \times 10^{-12}$; and hit trials versus miss trials, $p = 6.22 \times 10^{-34}$, and before cue onset group, hit trials versus baseline, $p = 1.81 \times 10^{-18}$; miss trials versus baseline, $p = 7.87 \times 10^{-8}$; and hit trials versus miss trials, $p = 5.22 \times 10^{-16}$. Mean $\pm$ SD.

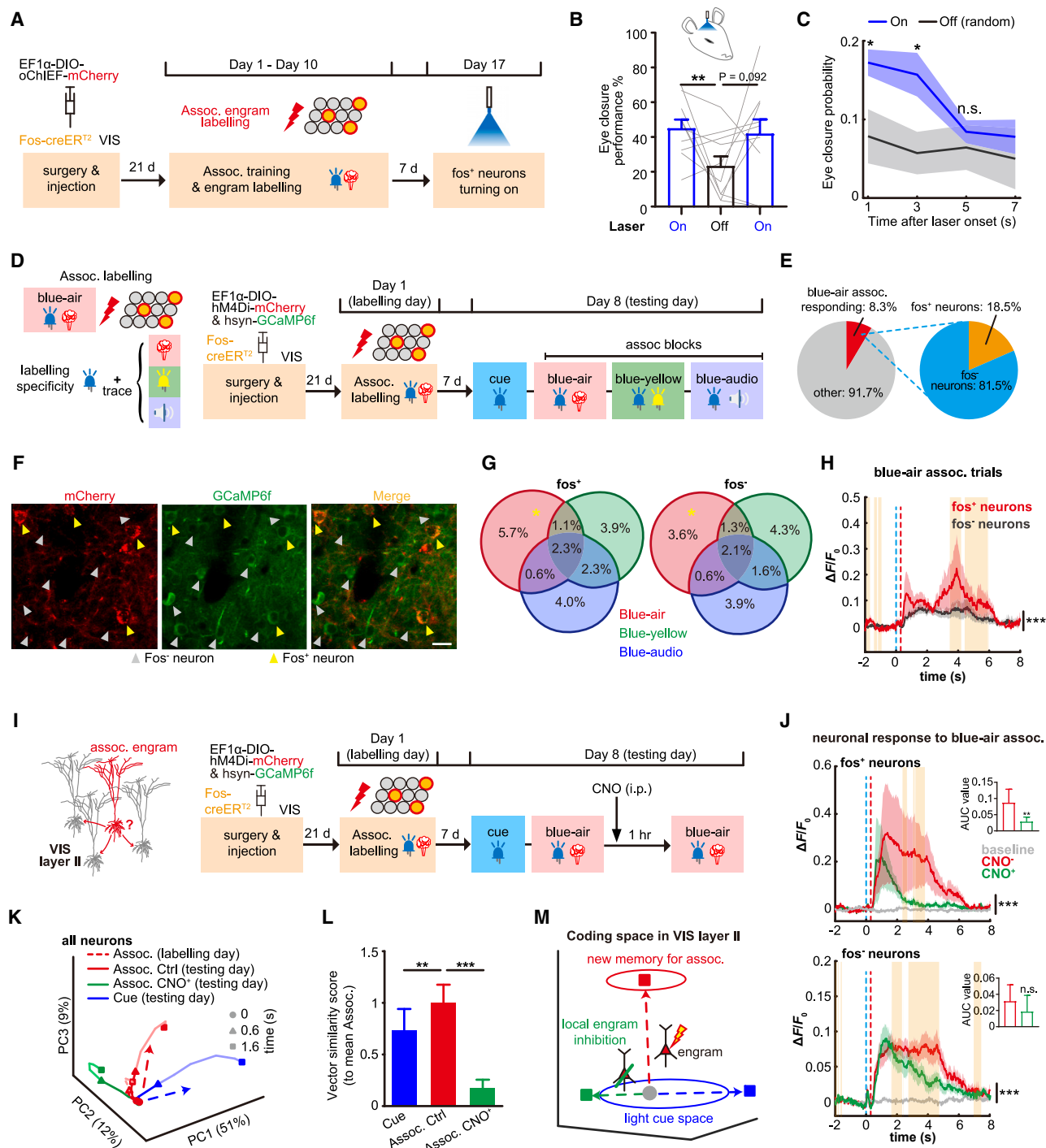


Figure 4. Fos⁺ cells are the hub neurons of the altered visual representation during association training

(A) The experimental design for labeling and activation of the fos⁺ neurons during 250 ms delay trace eyeblink conditioning with oChIEF-mCherry. (B) Eyelid closure rate. Paired t test: $n = 10$ mice, on 1st versus off, $p = 0.0038$ and on 2nd versus off, $p = 0.0920$. Mean $\pm$ SEM. (C) Eyelid closure probability distribution during laser-on or laser-off epochs. In laser-off epochs, the eyelid traces were randomly divided to segments with equal lengths of the laser-on trials for comparison. Paired t test: $n = 10$ mice, 0–2 s, $p = 0.0106$; 2–4 s, $p = 0.0125$; and 4–6 s, $p = 0.414$. Mean $\pm$ SEM. (D) Schematic and experimental design for labeling of the associative responding fos⁺ neurons through TRAP. The mice were exerted on only the blue-air association trials on the labeling day, and the mice were exerted on the blue-air, blue-yellow, and blue-audio trials in turn. (E) The pie chart shows the ratio of the fos⁺ cells in the blue-air-association responding neurons with GCaMP6f signals. (F) Images of TRAP-labeled fos⁺ neurons in layer II of VIS. Scale bar: 20 μ m.

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day (Figures S8F and S8G), suggesting that TRAP preferentially labels neurons with persistent activity.

To assess the fos-labeled neurons' role in maintaining the altered visual representations in the early phase of associative training, we inhibited the fos⁺ neurons. The fos⁺ neurons showed a significant reduction of activities in association blocks after CNO inhibition (Figures 4I and 4J). Moreover, the inhibition caused the cascading suppression of fos[−] neuronal activity, particularly in late-phase responses (Figures 4J and S8H). These findings indicate that TRAP-labeled neurons act as hubs supporting early-stage altered visual representations. Additionally, CNO inhibition preferentially affected blue-air association representations (for which TRAP neurons were labeled) but not the other pairings (Figures S8I–S8K), demonstrating functional specificity.

At the population level, the PCA in the control mice showed that association-induced visual representations on day 8 closely resembled those during neuronal labeling, with trajectories orthogonal to cue trials (Figures 4K, S8L, S9A, and S9B). In contrast, CNO-mediated inhibition of TRAP-labeled neurons altered the trajectories of the association blocks (Figure 4L), indicating disruption of the association-induced cortical representation. This inhibition also eliminated long-lasting Ca²⁺ activity in VIS following blue-air pairing, even in fos[−] neurons (Figures S9C and S9D). These findings demonstrate that TRAP-labeled neurons (engrams, presumably), although a minority within the neuron ensembles, could be the hub for sustaining the altered visual representations in association trials at the early stage of learning (Figure 4M).

VIP⁺ neurons in local circuit contribute to the altered visual representation induced by associative learning

To investigate how the association-evoked neuronal ensembles are chosen to emerge in VIS, we analyzed local inhibitory interneurons,^{27–32} including PV⁺, SOM⁺, and VIP⁺ neurons (Figure 5A). Injecting AAV9-hsyn-GCaMP6f and AAV9-DIO-EF1α-hM4Di-mCherry into VIS of PV-cre, SOM-cre, or VIP-cre mice (Figures 5B–5D and S10A), we assessed their responses to isolated or paired stimuli. PV⁺ neurons exhibited immediate activity suppression upon blue light, air puff, or paired stimuli (Figure S10B), suggesting the inhibitory input modulation role in sensory processing. SOM⁺ neurons displayed weak, variable responses across stimuli (Figure S10D), indicating minimal involvement in initial sensory encoding. In contrast, VIP⁺ neurons showed stimulus-dependent activation (Figure 5C), with individ-

ual cells exhibiting preferential response profiles in selectivity maps (see STAR Methods for details; Figure S10H).

CNO-mediated inhibition of the VIP⁺ interneurons during association conditioning reduced activity in most layer II neurons responsive to the association trials (Figures 5E and 5F). At the same time, ~5.3% of previously non-responsive neurons exhibited increased activities (Figure 5E). The scatterplots revealed marked changes in associative cellular representation following VIP⁺ inhibition compared to PBS controls (Figure 5G). In contrast, PV⁺ interneuron inhibition abolished selectivity across layer II neurons, triggering widespread activation (Figures S10C and S10F), indicating that PV⁺ neurons constrain active ensemble sizes. SOM⁺ inhibition also altered association-activated populations (Figures S10E and S10G).

Furthermore, VIP⁺ interneuron inhibition reoriented population activity trajectories away from both cue-only and intact association-evoked patterns (Figures 5H and S10I), indicating the structural reorganization of associative ensembles. The vector similarity analyses confirmed representational shifts between control and inhibition conditions (Figure 5I). As cortical interneurons integrate long-range inputs from multiple brain areas, i.e., basal ganglia, RSP, and lateral entorhinal cortex (LEC),^{23,33–36} they are subjected to the modulation of top-down regulations. Overall, our observations identified the altered visual representation in light-air association learning (Figure 5J). The altered visual representation, generated by hippocampal influences on local inhibitory circuits, is closely linked to associative memory formation, which does not occur without such visual representation changes.

DISCUSSION

While memory traces to encode association have been identified in various brain regions, including HPC, amygdala, and cortical areas,^{9,37–40} their generation and transformation mechanisms remain unclear. Using *in vivo* Ca²⁺ imaging, we observed light-evoked neuronal representations in layer II of mouse VIS being altered during early associative learning by the light-air puff pairing, forming nascent memory traces. These early visual representation changes require DG activity and the local inhibitory network, especially VIP⁺ neurons. Our findings emphasize critical HPC-cortex interactions during memory encoding, supporting HPC's indexing role in memory formation. Memory engrams emerge as a cortical subpopulation showing early associative activation and sustained activities after the pairing event. These engrams function as network hubs in the altered visual

(G) The Venn charts show the overlapping of the blue-air responding, blue-yellow responding, and blue-audio responding neurons in the fos⁺ and fos[−] ensembles ($n = 176$ for the fos⁺ neurons and 853 for the fos[−] neurons from 4 mice) on the testing day.

(H) The responding intensities of the blue-air-association responding neurons on the testing day. Yellow shading: time windows with significant differences (Mann-Whitney U test, $p < 0.05$). Two-way ANOVA test for the comparisons between the averaged curves: $p = 3.78 \times 10^{-125}$. Mean $\pm$ SEM.

(I) Schematic and experimental design for labeling and silencing the fos⁺ neurons through chemogenetics.

(J) The responding intensities of the blue-air-association responding fos⁺ and fos[−] neurons before and after Fos inhibition. Yellow shading: time windows with significant differences (Mann-Whitney U test, $p < 0.05$). Two-way ANOVA test for the comparisons between the averaged curves: fos⁺ ensemble, $p = 2.98 \times 10^{-128}$ and fos[−] ensemble, $p = 3.34 \times 10^{-96}$. Mean $\pm$ SEM. Welch's t tests for the bar plot: fos⁺, $p = 0.00166$ and fos[−], $p = 0.168$. Mean $\pm$ SD.

(K) The PCA trajectories of neuronal responses pre-/post-CNO-mediated Fos inhibition ($n = 4$ mice).

(L) The vector similarity score to the mean cue vector among three stimuli, including cue and association before and after CNO-mediated silencing of fos⁺ neurons on the testing day. Welch's t tests for the bar plot: cue versus association [assoc.] Ctrl, $p = 0.0210$; cue versus assoc. CNO⁺, $p = 6.65 \times 10^{-6}$; and assoc. Ctrl versus assoc. CNO⁺, $p = 4.29 \times 10^{-4}$. Mean $\pm$ SD.

(M) The working model of fos⁺ neurons or engram cells in layer II of VIS in the new association memory formation.

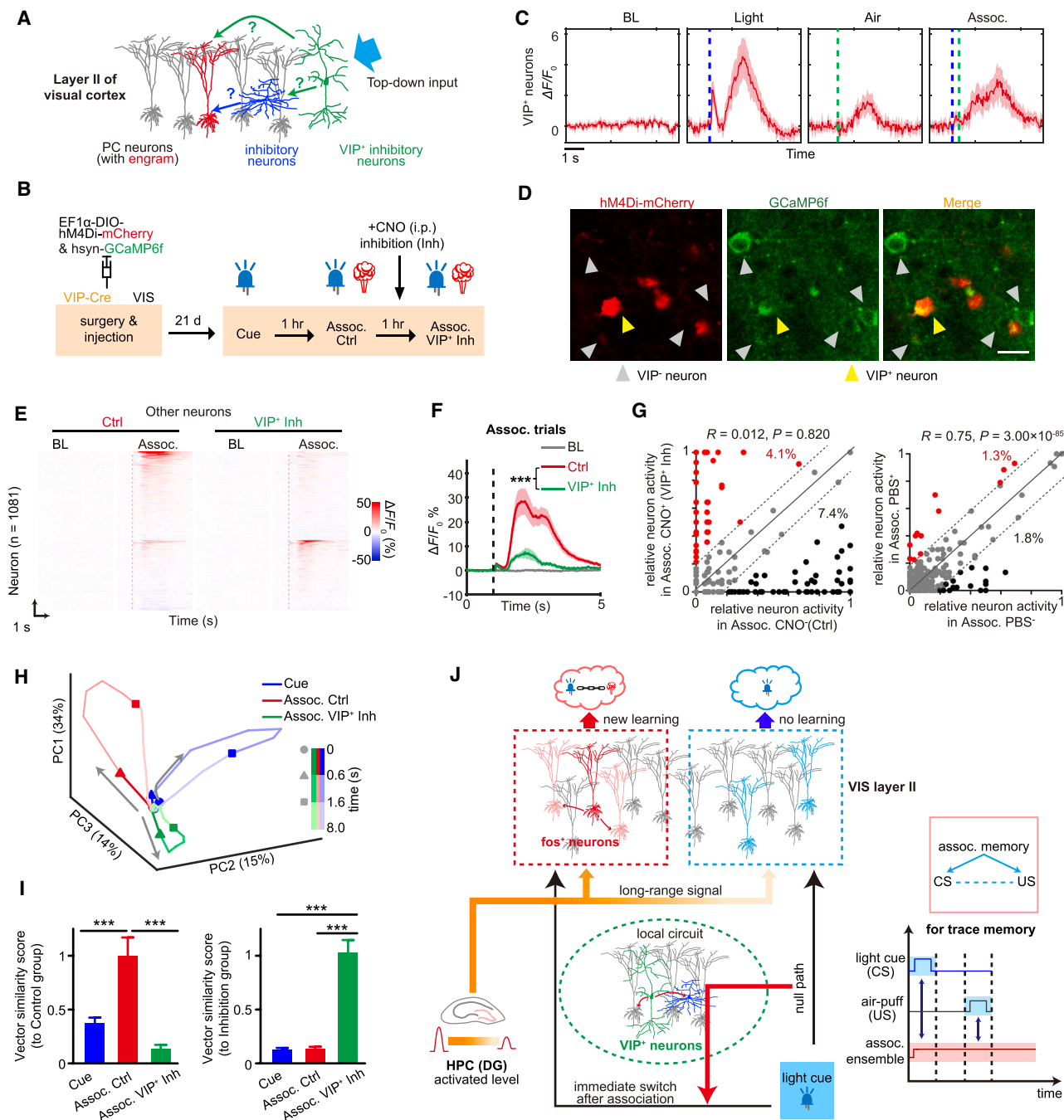


Figure 5. VIP+ neurons in local circuit modulate neuronal representation for associative memory in VIS

(A) Local circuits involved with different interneurons.
(B) Schematic and experimental design for labeling of the association neurons with mCherry through TRAP and silencing the VIP+ neurons through chemogenetics.
(C) Traces of averaged calcium activities VIP+ ($n = 50$ cells from 4 animals) interneurons to blue light, air puff, or association stimuli. Dashed line, onset of blue light (blue) or air puff (green). Mean $\pm$ SEM.
(D) Images of TRAP-labeled VIP+ neurons in layer II of VIS. Scale bar: 20 μ m.
(E) Color raster plot shows VIP+ neurons before and after CNO-mediated silencing of VIP+ interneurons. Dashed line: onset of stimulus.
(F) Traces of averaged calcium activities of association neurons before and after CNO-mediated silencing of VIP+ interneurons (two-way ANOVA, $p = 1.69 \times 10^{-8}$). Mean $\pm$ SEM.

(legend continued on next page)

representations, encoding new associative memories bound to visual cues.

Hippocampal-cortical interactions are crucial for memory formation and retrieval.⁴¹ Standard consolidation theory posits that HPC guides cortical module integration over time, with remote memories becoming HPC-independent once stable inter-cortical connections are formed.^{42–46} In this work, we revealed immediate visual representation changes in VIS after associative learning, indicating early cortical memory trace formation during encoding. These emerging memory traces are dependent on activities in HPC (DG), indicating the direct involvement of the HPC circuit as the index in the establishment of cortical memory traces. This mechanism may facilitate hippocampal reactivation of cortical representations during retrieval.⁴⁷ Our findings establish that hippocampal signals actively drive cortical memory trace initiation, providing empirical support for the hippocampal indexing theory. While memory traces form in distinct regions of neocortex during learning, hippocampal-cortical circuits remain dynamically engaged across both encoding and post-learning phases, revealing spatiotemporal coordination in cortical memory network maturation.

Hippocampal-cortical interactions depend on long-range projections.^{45,48} We suggest that the classical hippocampal output pathways and EC circuits may relay DG activity to VIS. LEC, a critical hippocampal-cortical relay, plays a key role in memory encoding and storage.^{11,49,50} Our previous work demonstrated that EC-cortical connectivity modulates learning-induced gamma synchrony across cortices during spatial and contextual fear memory tasks.²³ Oscillatory synchrony, regulated by local inhibitory neurons, is an essential modulator of information processing in the brain.^{51–54} We identified VIP⁺ interneurons selecting the altered cortical representations, suggesting the local inhibition circuit plays a key role in the long-range interaction to generate memory traces in VIS. The involvement of the local inhibitory network may confer flexibility and dynamics for memory encoding and retrieval.⁵⁵

Memory engrams are traditionally labeled using genetic methods with relatively low temporal resolution.⁵⁶ We observed real-time VIS activities during learning with TRAP labeling. The engrams consist of a small part of the learning-engaged neurons. Importantly, engram cells showed sustained activity long after stimuli ceased in early learning. The higher and prolonged neuronal activities of the engrams suggest pre-conditioning of memory traces and memory allocation.^{57–59} Additionally, the engram activities are essential for the long-lasting later phase of association-induced neuronal representation, even for the fos[−] population, indicating engrams act as central hubs for

persistent associative memory traces in VIS during memory encoding.

Furthermore, as engrams showed distinct physical/chemical features at the early phase of learning, this strongly suggests a sub-network in the brain in a priming state to incorporate upcoming events for memory encoding and subsequent consolidation processes. Consistent with memory allocation principles and hippocampal place cell preplay phenomena,^{60,61} our discovery suggests an alternative for the traditional view of memory formation as a passive process, hypothesized by Hebb's rule. It highlights an active brain network subset prepared for memory encoding. Moreover, our studies discover that in mammalian brain, the active component of the network is selected by internal signals, coming from HPC, to encode associative memories.

Generally, our findings demonstrate hippocampus-driven visual cortical reorganization during associative learning's initial phase. The single-neuron *in vivo* imaging reveals dynamic activity patterns and structural network configurations underlying memory trace formation. This discovery paves the way for future functional dissection of cortical-hippocampal interaction on memory encoding.

Limitations of the study

We mentioned that the eyeblink trace conditioning has some shortages in the investigation of associative memory itself. Firstly, CR-US coupling causes overlapping neural responses to the CS, US, and UR, confounded by sensory-motor interference. Secondly, Fos-based engram labeling lacks sufficient specificity due to low temporal resolution. Thirdly, we are unable to provide more precise spike timing information for the neuronal population because of the kinetic properties of calcium fluorescence imaging. Finally, the physiological linkages to regulate the cortical-hippocampal crosstalk during the associative memory trace formation remain unclear. Our previous work may provide some hints to the answer,²³ but more detailed investigations are needed.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Ji-Song Guan (guanjs@shanghaitech.edu.cn).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- The necessary codes for the analysis used in this paper have been deposited at Zenodo at <https://doi.org/10.5281/zenodo.15023876> and

(G) Left: scatterplot of response probability of association neurons before and after CNO-mediated inhibition of VIP⁺ interneurons ($n = 1,081$ neurons from 4 mice). Right: scatterplot of response probability of association neurons before and after delivery of PBS ($n = 894$ neurons from 4 mice). Red dots: the neurons that showed enhanced activities after manipulation. Black dots: the neurons that showed reduced activities after manipulation. Dashed line, 99% confidence interval.

(H) The PCA trajectories of the neuron ensemble in response to the cue trials and the association trials before and after CNO-mediated silencing of VIP⁺ neurons in layer II of VIS on the testing day. The gray arrows indicate the direction of the trajectories.

(I) The vector similarity score to the mean association before (left) and after (right) silencing vector on the testing day. Welch's *t* tests for the bar plot: left, cue versus assoc. Ctrl, $p = 1.01 \times 10^{-9}$; cue versus assoc. VIP⁺ inhibition (Inh), $p = 2.30 \times 10^{-7}$; and assoc. Ctrl versus assoc. VIP⁺ Inh, $p = 3.84 \times 10^{-13}$, and right, cue versus assoc. Ctrl, $p = 0.0133$; cue versus assoc. VIP⁺ Inh, $p = 1.94 \times 10^{-11}$; and assoc. Ctrl versus assoc. VIP⁺ Inh, $p = 2.24 \times 10^{-12}$. Mean $\pm$ SD.

(J) The working model of VIP⁺ interneurons in the association pattern formation in layer II of VIS.

<https://doi.org/10.5281/zenodo.15023816> and are publicly available as of the date of publication.

- Any raw data or tools that support the findings of this study are available upon reasonable request to the lead corresponding author (J.-S.G.).

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AUTHOR CONTRIBUTIONS

J.-S.G. conceived and designed the project. H.X. developed the methodology and planned the project. C.L. and Z.L. performed the experiments. K.L. performed data analysis. T.Z. helped with data analysis. G.W. helped with the experiments. All the authors commented on the manuscript.

DECLARATION OF INTERESTS

The authors declare that they have no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
AAV9-hsyn-GCaMP6f-WPRE	Obio Technology	N/A
AAV9-EF1 α -DIO-hM4Di-mCherry-WPRE	Obio Technology	N/A
AAV9-EF1 α -DIO-oChIEF-mCherry	Obio Technology	N/A
AAV9-hsyn-oChIEF-EYFP	Obio Technology	N/A
AAV9-hsyn-EGFP	Obio Technology	N/A
Experimental models: Organisms/strains		
Mouse: C57BL/6J	Shanghai Jihui Lab Animal Care	RRID:IMSR_JAX:000664
Mouse: Fos-creER ^{T2}	The Jackson Laboratory	RRID:IMSR_JAX:021882
Mouse: PV-cre	The Jackson Laboratory	RRID:IMSR_JAX:017320
Mouse: VIP-cre	The Jackson Laboratory	RRID:IMSR_JAX:010908
Mouse: Som-cre	The Jackson Laboratory	RRID:IMSR_JAX:013044
Chemicals, peptides, and recombinant proteins		
Chlortetracycline Hydrochloride Eye Ointment	HuaQing pharmaceutical	N/A
Povidone-iodine	Lircon Medical Technology	N/A
Denture Base Polymer	NISSIN	N/A
Isoflurane	RWD	R510-22-10
D-PBS (without Ca ²⁺ or Mg ²⁺)	Beyotime	C0221D
PBS	Beyotime	C0221A
DAPI	Beyotime	C1002
Ethanol 75%	Sinopharm Chemical Reagent (Shanghai, China)	80176961
Continued		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
Ethanol absolute	Sinopharm Chemical Reagent (Shanghai, China)	10009228
Dimethyl Sulfoxide	Accela	SY006600
Dexamethasone	Abcam	ab120743
Corn oil	AbMole Bioscience	M9109
Paraformaldehyde	Sigma-Aldrich	158127
Clozapine-N-oxide	Sigma-Aldrich	C0832
Tamoxifen	Sigma-Aldrich	T5648
Software and algorithms		
GraphPad Prism 8.0.0	GraphPad Software	N/A
MATLAB R2021a	MathWorks	N/A
ImageJ	https://imagej.nih.gov/ij	N/A
Calcium signal processing code	This paper	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Mice

The animal care practices at National Facility for Protein Science Shanghai (NFPS) were in accordance with institutional guidelines and approved by the Institutional Animal Care and Use Committee (IACUC) at ShanghaiTech University. All animals used were male and were housed in groups of 3–5 on a standard 12-h light/12-h dark cycle with food and water free available. Fos-creER^{T2}, PV-cre, VIP-cre and SOM-cre mouse were distributed from Jackson Laboratories. WT mice were obtained from Shanghai Jihui Laboratory Animal Care Co.,Ltd and kept in Shanghai Lab Animal Research Center (SLARC). Male mice of 15-16-week-old were used in the calcium imaging experiments. The animals were returned to the homepage after daily training or experiments.

METHOD DETAILS

Behavioral tasks

A Behavior experiments were performed in a custom-designed, sound-attenuated dark box. The mice were head-fixed with a custom-designed frame made of sandblasted 7075 aluminum alloy. The frame was stably mounted to the imaging platform or behavior box using clamps and thumbscrews. A blue LED (460 nm), yellow LED (590 nm), a DC buzzer (3 kHz, 70 dB), and an air-puff delivery tube were positioned around 5 cm away from the mouse's eye, opposite the imaged region. The timing and control of these elements were managed through custom-written programs using a microcontroller (Arduino UNO). The mice were handled and habituated to the training environment and head fixation for up to 45 min a day for at least two days prior to training or imaging.

For the blue-air trace conditioning, the daily training included pseudo-randomly interleaved association trials (80%) and cue (a.A.) trials (20%). In the association trials, the stimuli were blue-light-trace-air-puff, consisting of 50 ms blue-light, 250 ms delayed-trace and 20 ms air puff, with 10 ± 2 s random inter-trial interval. In cue (a.A.) trials, we used the same parameters in association trials but no air puff was delivered. Mice underwent ~60 trials of training per day for 6–14 days and eventually learned to close their eyes during the 300-ms responding window (50 ms cue ongoing +250 ms delay). Correct responses were defined as detectable eyelid blinking during delay. The correct rate reached >50% in 7–10 days in most mice. For the TRAP experiment, mice were light-deprived in a dark cage for two days before and after labeling to reduce non-specificity. On the labeling day (24 h post TM injection), more than 60 trials of association trials were performed with no recall trials. Fos⁺ neurons were detected ~2 weeks after labeling.

For the other types of sequential stimulations, the mice were also head-fixed in the same chamber used in the blue-air trace conditioning. The light stimulus was delivered with either blue or yellow LED, lasting 50 ms for each trial. The air puff and audio were delivered as described as above, lasting 20 ms and 50 ms respectively. For each mouse, 12 trials of light stimuli were delivered first. Then, 12 trials of air puff or audio stimuli followed. The last block was the association of light and air puff or audio for 12 trials as well. Between each block of stimulus, there was 10 min resting. The association was 50 ms light-250 ms delay-20 ms air puff or 50 ms audio.

We used a high-speed camera (>30 fps) to record eyelid movements in mice. A CR is counted as the mouse started to close its eyelids within the trace temporal window between the light and the air-puff both in the 250 ms and 15 s trace conditioning. For each mouse, the “eye closure CR (%)” is the ratio of the trials with a CR in all the association trials. The “normalized eyeball exposed (%)” is the extent of eyelid closure of the mouse. We used the parameter to describe the mouse eyelid closure movement in the experiments. The value of “normalized eyeball exposed (%)” was also measured with the high-speed camera video. We define the complete eyelid closure (no eyeball areas were recorded in the video) as the “100% eye closure” and the spontaneous eye opening status of the mouse (largest eyeball areas were recorded in the video) in the experimental device as the “0% eye closure”. The areas of the eyeballs recorded during the experiments were measured and the “normalized eyeball exposed (%)” was calculated using the linear fitting between the 0% and 100%.

Mouse surgery with virus injection and fiber implantation

All surgeries were performed on 10–12-week-old mice using stereotaxic guidance. The mice were anesthetized with 5% isoflurane in O₂ and immobilized in a custom-built stage-mounted stereotaxic apparatus (Thinkerbiotech Co., Ltd). A midline incision was made and the scalp was reflected to expose the skull.

For the imaging experiments, a circular craniotomy (4–5 mm in diameter) was created using a high-speed drill and the dura was removed in the target regions. The viruses used in this study, AAV9-hsyn-GCaMP6f, AAV9-EF1 α -DIO-oChIEF-mCherry, AAV9-EF1 α -DIO-hM4D-mCherry, AAV9-hsyn-oChIEF-mCherry and AAV9-hsyn-EGFP (titers varied from 0.5 to 2×10^{13} genomes ml⁻¹) were bought from Obio Technology (Shanghai) Co., Ltd. A pulled glass pipette (Wiretrol II 5 and 10, Drummond) was beveled at 45° and filled with mineral oil, then connected to a custom-built manually-operated hydraulic pump (Thinkerbiotech Co., Ltd) for virus injection. The volume of injection was approximately 100 nL, and the speed was approximately 50 nL/min. The pipette was carefully positioned around the target sites (VIS: AP -2.7 mm, ML 2.5 mm, DV -0.3 mm from the bregma, the same hereafter; DG: AP -2.0 mm, ML $\pm$ 1.5 mm, DV -2.0 mm for the experiments in Figure 2K; RSP: AP -3.0 mm, ML $\pm$ 0.8 mm, DV -0.2 mm; PTLp: AP -2.3 mm, ML $\pm$ 2.0 mm, DV -0.3 mm; mPFC: AP 1.76 mm, ML $\pm$ 0.36 mm, DV -0.2 mm, DV 2.8 mm) and remained in place for 2 min before the injection began. Multiple injections were performed at various points around the target site, with distances of at least 1 mm between them. After the injections, the pipette was left in place for an additional 2 min before being slowly withdrawn. A cranial window and custom-made titanium head-plate (Fig. S1) were implanted to the target region and secured in place with denture base polymer (NISSIN). Mice were allowed to recover for approximately 30 days before any behavior tasks or imaging experiments were performed.

For the optogenetic experiments, the fibers (d = 200 μ m, N.A. = 0.48, Thorlabs) were implanted and secured by black base polymer. Implants were placed bilaterally in the DG (AP -2.0 mm, ML $\pm$ 1.5 mm, DV -2.0 mm) and VIS (AP -2.7 mm, ML 2.5 mm, DV -0.3 mm) in the optogenetic activation experiments. For the experiments with both microscopy imaging and optogenetic activation in Figure 2K, the fibers were implanted with an angle of 45° to the skull at the location (AP -2.0 mm, ML 3.2 mm, DV -2.4 mm).

In vivo two-photon imaging

3–5 months old mice received virus injection and cranial window implantation as previously described. Imaging began 1 month after surgery. Mice were anesthetized with 5% isoflurane (in O₂) for a while and placed into a custom-built frame for imaging. Mice were head-fixed and awake during imaging.

Calcium signals were acquired using an Olympus FVMPE-RS upright microscope equipped with pre-chirp optics and a fast AOM coupled with a 2 mm working distance, 25× water immersion lens (numerical aperture, NA 1.05). A mode-locked titanium/sapphire laser generated two-photon excitation at 960/1,100 nm and four photomultiplier tubes collected emitted light in the range of 380–480, 500–540, and 560–650 nm. The laser output power was maintained at 2.5 W and the power reaching the mouse brain was around 10 mW, and it was kept consistent during time-lapse imaging. The field of view was 509 × 509 μm. The target regions were imaged at ~30 frames/sec (one-layer scanning) or 8 frames/sec (two-layer scanning) for a duration of 120 s (for 30 frames/sec trials) or 150 s (for 8 frames/sec trials) as one imaging block. There was an interval of no less than 30 s between two blocks. The recording and stimulation were initiated at the same time through a custom-made controller. Synchronization between the images and the stimulation was achieved through stimulation-synchronized tag signals sent to the microscope (Figure S1).

Optogenetic manipulation

In the optogenetic tests, no sensory stimuli were presented. During the laser-on trials, a 473 nm laser (from Lasever Inc.) was delivered to the targeted sites in the visual cortex through implanted fibers. The laser power at the fiber's terminal was calibrated and maintained at approximately 10 mW during each session. To prevent any potential disturbance to the mouse's eyes from the laser illumination, black nail polish mixed with cyanoacrylate glue was used to cover the potential path of the laser. This ensured that there was no visible difference between the laser-on trials and laser-off trials as viewed by the mice. Eye closure within the response window after laser onset were recorded by a high-speed camera movie. During the laser-off trials, spontaneous eye closure was randomly divided into segments of equal length for comparison with the laser-on trials. In the experiments with 15-s trace, we applied the 2 s optogenetic activation to avoid the temporal interference. The major concern was that the continuous DG stimulation until the US onset might cause DG activity as a cue itself without a long enough trace time, which contradicted the question of how the associative stimulation with temporal separation is represented and regulated. Therefore, in the 15 s trace paradigm, we only introduced DG activation around the CS for 2 s, maintaining a 14 s blank trace before the US onset, to make the result more convincing. Originally, we had tried different stimulus durations from 2 s up to 13 s (1 s before CS onset and the remaining duration after the CS onset) and found that the DG activation, even if only in the early stage (2 s), still yielded significant behavior changes. Further processing of the behavior data was performed using custom-written code in MATLAB.

Drug injections

We used Tamoxifen (TM; Sigma-Aldrich) to labeled active neurons within a targeted time window after TM injection i.p. TM was made in a solution of 20 mg/mL in corn oil and injected (i.p.) at the dose of 150 mg/kg 24 h before traced eyeblink conditioning training.

We used Clozapine-N-oxide (CNO; Sigma-Aldrich) to inhibit hM4Di labeled neurons/inter-neurons. CNO was made in a solution of 0.6 mg/mL in PBS and injected (i.p.) at the dose of 2 mg/kg 60 min before the imaging or behavior experiments.

Corn oil (AbMole) and PBS (Beyotime) were used as the control vehicle to TM and CNO respectively.

QUANTIFICATION AND STATISTICAL ANALYSIS

Image process and analysis

Data processing, which included picture alignment and data acquisition, was conducted using custom-written semi-automatic MATLAB codes. The tag channel, indicating the start of sensory stimulation, was first separated from the imaging data. Picture alignment was then performed on the GCaMP6f channel to correct for brain motion caused by animal movement. Each frame of the first baseline trial was aligned to the first frame based on the maximum correlation coefficient, with a maximum shift of 50 pixels in both the x and y axis, between the translated images and the reference image. All frame of the first baseline trial were then averaged to create a high-quality reference image. Subsequent trials were aligned with this reference image using the same algorithm. To align images captured on different days, both translation and rotation were performed to correct for brain motion and head fixation angle deviation.

To extract the fluorescence signals from each neuron, we manually labeled the GCaMP6f⁺ neurons using ImageJ. We used the mean, maximum intensity and standard deviation of all frames in a trial to determine region of interest (ROI) for each GCaMP6f⁺ neuron. On average, we obtained around 200 GCaMP6f⁺ neurons per field of view. The fluorescence signal of each neuron was calculated as the mean of all pixels within its ROI. To eliminate the influence of the diffused light on cellular signals,⁶² we normalized the fluorescence signal of each neuron by subtracting its background signal. The background signal of each neuron at each frame was determined as the mean of half of the pixels located 20 μm outside the neuron with relative low fluorescence signals.

To analyze the calcium levels in each neuron, the calcium activity of each neuron was divided into 8–10 s segments based on the timing window to observe the changes in calcium activity. The calcium activity of each neuron for each trial (*F*) was normalized by its averaged baseline activity (*F*₀) to calculate the ratio of calcium signal changes $\Delta F/F\% = (F - F_0)/F_0 \times 100$.

Since PV⁺ interneurons had strong spontaneous activities and were inhibited by stimulation, the normalization of calcium traces of PV⁺ interneurons was different from the other neurons. Instead of using the mean activity of a 2-s baseline period before stimulus

onset, the normalization was done using the mean activity of a 2-min recording of spontaneous calcium activities (F_0) without any additional sensory input (referred to as baseline trial).⁶³ This baseline trial was randomly divided into multiple 10-s segments to match the segmented calcium traces during stimuli, and the same normalization process was applied.

Principal component analysis

The data was analyzed using principal component analysis (PCA) to show the main activity pattern of the neuron ensemble. The PCA was performed on the calcium data from the onset of stimuli to 8 s after stimuli. The data was treated as a whole vector for each moment in time, with all neurons from all mice considered together. All of the time points from each stimulus were used to calculate the first three principal components as the main coding space. All of the data was processed by the MATLAB function `pca`. To smooth out some single-experiment errors, the mean data of several trials were used. Considering meaningful calcium events are positive, the positive direction of a principal component in calcium imaging represents one activity pattern, while the negative direction represents a different activity pattern. Thus, we only count the sum of positive weights in Figure S1.

Vector similarity score

The parameter of vector similarity score measures the similarity between different types of stimuli accurately by giving an exact measurement of similarity between vectors. This measurement is intuitively understood as the inner product between two normalized vectors (Fig. S3b). To obtain more stable data, the 1-s activity among the peak is first summed to obtain area under the curve⁶³ for a specific time period. By normalizing the vectors to unit vectors, we eliminate the effects of extreme values and only focus on the direction of the vectors. Next, we calculated the mean vector of the basal type of stimuli $\bar{\vec{b}} / \|\bar{\vec{b}}\|$ as the main direction. For example, the association stimulus was the basal stimulus of Figure 11 so that the mean vector of the association stimulus would be picked. For each trial of a specific stimulus, we calculated the inner product $U_{a,b}$ between the direction from one trial $\vec{a}_i / \|\vec{a}_i\|$ and the main direction $\bar{\vec{b}} / \|\bar{\vec{b}}\|$. As a comparison, the inner product $U_{a,a}$ between the direction from one trial $\vec{a}_i / \|\vec{a}_i\|$ and the main direction of itself $\bar{\vec{a}} / \|\bar{\vec{a}}\|$ from the same stimulus was calculated. For a specific trial, $\vec{a}_i$ and the basal type of stimuli $\bar{\vec{b}}$, the vector similarity score $S_{a,b}$ is defined as:

$$S_{a,b} = U_{a,b} / \frac{1}{n} \sum_{i=1}^n U_{a,a}.$$

Then the vector similarity score S_{ab} is defined as:

$$S_{ab} = \frac{1}{n} \sum_{i=1}^n S_{a,b}.$$

Here n is the number of trials of stimulus a .

For a simple proof, we will show that $S_{ab} \leq 1$ and $S_{ab} = 1$ if and only if $\bar{\vec{b}} = \bar{\vec{a}}$.

$$\begin{aligned} S_{ab} &= \frac{\frac{1}{n} \sum_{i=1}^n U_{a,b}}{\frac{1}{n} \sum_{i=1}^n U_{a,a}} \\ &= \frac{\frac{1}{n} \sum_{i=1}^n (\vec{a}_i \cdot \bar{\vec{b}} / [\|\vec{a}_i\| \cdot \|\bar{\vec{b}}\|])}{\frac{1}{n} \sum_{i=1}^n (\vec{a}_i \cdot \bar{\vec{a}} / [\|\vec{a}_i\| \cdot \|\bar{\vec{a}}\|])} \\ &= \frac{\left(\frac{1}{n} \sum_{i=1}^n [\vec{a}_i / \|\vec{a}_i\|] \right) \cdot \bar{\vec{b}} / \|\bar{\vec{b}}\|}{\left(\frac{1}{n} \sum_{i=1}^n [\vec{a}_i / \|\vec{a}_i\|] \right) \cdot \bar{\vec{a}} / \|\bar{\vec{a}}\|} \\ &= \frac{\bar{\vec{a}} \cdot \bar{\vec{b}} / \|\bar{\vec{b}}\|}{\bar{\vec{a}} \cdot \bar{\vec{a}} / \|\bar{\vec{a}}\|} \end{aligned}$$

$$= \frac{\overline{\vec{a}} \cdot \overline{\vec{b}}}{\|\overline{\vec{a}}\| \cdot \|\overline{\vec{b}}\|}.$$

So $S_{ab} \leq 1$ and $S_{ab} = 1$ if and only if $\overline{\vec{b}} = \overline{\vec{a}}$.

In the actual calculation process, when the number of samples was unbalanced, we used down sampling to calculate $\overline{\vec{a}}$ for each $\vec{a}_i$. Nevertheless, the above proof guarantees that the vector similarity score is always between -1 and 1 , excluding some statistical noise disturbances. Besides, we used vectors from other trials when $\vec{a}_i$ and $\vec{b}$ belongs to the same type of stimuli to correct for the bias caused by sampling. The above calculation process was done by custom-written codes in MATLAB.

Vector intensity

Similar to the vector similarity score, the vector intensity is defined as the inner product between the raw vector $\vec{a}_i$ (without normalization) and the main direction $\overline{\vec{b}}/\|\overline{\vec{b}}\|$ and is calculated by custom-written codes in MATLAB (Fig. S3c). That is

$$I_{a,b} = \vec{a}_i \cdot \frac{\overline{\vec{b}}}{\|\overline{\vec{b}}\|}.$$

Then I_{ab} is defined as:

$$I_{ab} = \frac{1}{n} \sum_{i=1}^n I_{a,b}.$$

This paradigm I_{ab} represents the activity power of b pattern in a component.

For the second condition, we only calculated the vector intensity score when $\vec{a}_i$ and $\overline{\vec{b}}$ belonged to the same type of stimuli. Same as the calculation process of vector similarity score, the $\overline{\vec{a}}$ would be replaced by the estimated value $\overline{\vec{a}_i}$ which was computed by several vectors other than $\vec{a}_i$.

Then I_a is defined as:

$$I_a = \frac{1}{n} \sum_{i=1}^n I_{a,a}.$$

In this case, the vector intensity score could reflect the activity pattern strength of a certain stable pattern but not simple mean activity strength.

Statistic methods

We took two steps to identify the neurons responding to different stimuli. We first set a threshold (T) for an active calcium event for each neuron based on its variation in the baseline trials. The threshold was calculated as $T = 2 \times \text{median}(\Delta F \text{ baseline}) - \text{minimum}(\Delta F \text{ baseline})$. For each neuron, we listed the maximum ΔF value within 3 s after the onset of the first cue (usually the blue-light flash) of every trial. If the listed value was larger than T , we would record an active trial for the corresponding neuron. Secondly, we calculated the ratio of active trials for each neuron across all blocks of each stimulation type (such as light cue, air-puff, and blue-light pairing) on the same day. The neurons with an active trial ratio 3.0 times higher than the average of all neurons were defined as the responding neuron to the stimulation. Since the population sizes of interneurons were relatively small (less than 100), the threshold for response probability was artificially set to 10%, estimated from the variation level of excitatory neurons.

To quantify the selectivity of responsive neurons to stimuli, we first identified all responsive neurons for each of the four stimuli (baseline, blue light, air puff, and association trials). These responsive neurons were then considered to have a significant stimulus selectivity if their response to each stimulus was significantly different from (0,0) according to a Hotelling's T-squared test ($p < 0.05$). To visualize this selectivity, we plotted the response of each neuron to each stimulus as a point in polar coordinates,⁶⁴ creating a selectivity map. In our experimental set-up, each selectivity map had four points, with polar angles assigned to baseline (0°), blue light (90°), air puff (180°), and association trials (360°), and the polar radius defined by the response intensity t of each neuron to each corresponding stimulus. The preferred stimulus for each selectively responsive neuron was determined as the stimulus to which the neuron had the strongest response.

GraphPad Prism version 8.00 (GraphPad Software, La Jolla, California, USA) and MATLAB R2021a were used for statistical analyses. Statistical significance was assessed by Wilcoxon signed rank test (for paired samples), Mann-Whitney U test (for unpaired samples), Friedman test followed by the Dunn's multiple comparisons test, two-way repeated ANOVA (for paired samples) followed by Holm-Sidak's test, two-way mixed ANOVA (for unpaired samples) followed by Holm-Sidak's test and t test where appropriate (see the figure legends for each test). Data collection and analysis were not performed blind to the conditions of the experiments, since all data were measured automated with no artificial intervention.